

Deepikarthini Kanche

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PROFESSIONAL SUMMARY

Computer Science Engineering graduate with hands-on experience in software automation, AI infrastructure validation, observability, and performance benchmarking. Skilled in QA automation, API testing, LLM deployment, and monitoring enterprise-scale systems using modern testing and observability tools. Strong analytical, debugging, and problem-solving abilities with exposure to Agile workflows and scalable infrastructure environments.

EXPERIENCE

Software Test Engineer Apprentice — UCS AI Sizer Team June 2025 - Present
Cisco Systems (India) Private Limited, Bengaluru

- **Deployed Mistral/Llama models via NVIDIA NIM containers on Cisco UCS C885A M8 servers (H200 NVL GPUs)**, executing high-concurrency AIPerf benchmarks to validate throughput, GPU utilization, TTFT, and p95/p99 latency metrics for mission-critical workloads.
- **Designed multi-turn conversational and mixed short/long-context simulations using Poisson traffic patterns**, monitoring production-level inference KPIs and VRAM telemetry utilizing NVIDIA DCGM exporters, Prometheus, and Grafana dashboards.
- **Automated shell scripting pipelines** to synthesize 500-session multi-agent workloads and stress-test a **Llama-3.3-70B model** under a 100-worker concurrency load on **NVIDIA H100 hardware**, while managing containerized data extraction workflows across Docker layers to capture critical system throughput and performance logs.
- **Optimized Triton/vLLM architecture efficiency** by auditing Key-Value (KV) cache block allocations to ensure zero block evictions under peak load, while troubleshooting and patching client-side JavaScript runtime scoping errors within interactive reporting dashboards to successfully restore live data visualizations.
- **Tech:** NVIDIA NIM, Triton Inference Server, vLLM, AIPerf, Llama-3.3-70B, Mistral, NVIDIA H100/H200 GPUs, Cisco UCS C885A M8, DCGM Exporter, Prometheus, Grafana, Docker, Linux Shell Scripting, Python, JavaScript.

EDUCATION

Sri Chaitanya Technical Campus, Hyderabad Aug 2020 - Jun 2024
Bachelor of Technology — Computer Science and Engineering CGPA: 7.58/10

Gouthami Junior College, Hyderabad Feb 2018 - Mar 2020
Class XII (TSBIE) Percentage: 94.30%

PROJECTS

Bug Reproduction and QA Workflow Management System

React, Spring Boot, PostgreSQL

- Developed a full-stack application to streamline bug reporting, reproduction tracking, assignment workflows, and status management for QA and development teams.
- Built RESTful APIs using Spring Boot and integrated PostgreSQL for efficient bug data storage and retrieval.
- Implemented role-based access workflows for testers, developers, and administrators to improve collaboration and defect lifecycle management.
- Designed responsive React components for real-time bug tracking, workflow updates, and issue monitoring.
- Improved debugging efficiency by enabling structured reproduction steps, environment tracking, and centralized issue management.

TraceOps — AI-Powered Root Cause Analysis System

Python, FastAPI, LangChain, ChromaDB, React.

- Developed a RAG-based incident analysis system using FastAPI, ChromaDB, and LLMs to automate root cause identification from application logs.
- Implemented semantic log retrieval with vector similarity search and OpenAI embeddings, enabling intelligent issue correlation across 15+ log patterns with sub-second response times.
- Built a React-based dashboard for incident investigation and AI-driven debugging, and deployed the solution using Render and Vercel.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, JavaScript, TypeScript, Spring Boot, FastAPI, React, Angular, HTML5, CSS3.

AI & Testing Automation: Machine Learning, AIPerf, LangChain, HuggingFace, ChromaDB (RAG), Selenium, Postman.

DevOps, Cloud & Databases: Docker, Git, GitHub, Jenkins, Grafana, Prometheus, Linux, Basics of AWS, MySQL, PostgreSQL.

Core Computer Science: OOP, DBMS, Operating Systems, Computer Networks, LLM Engineering.

CERTIFICATIONS & ACHIEVEMENTS

Cisco Certified Network Associate (CCNA) — Cisco Jun 2026
- **Problem Solver:** Solved problems on Leetcode and GFG, strengthening DSA and competitive programming skills. Ongoing